Diagnose, repair, recheck

antocciphysics | TEACHER KEY | CPA

Learning target: Identify a reasoning error and demonstrate the repaired method on a new problem.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 16; 3 course days later.*
Net 21 N right; acceleration 7 m/s² right. Leftward motion initially slows.

02. Explain a worked example (worked)
Directions match while speeding up and oppose while slowing in straight-line motion; motion direction alone is insufficient.

03. Position is not velocity (model)
Position change 6 m left in 3 s; velocity 2 m/s left.

04. Net force is not one force (practice)
Subtract opposing force first: 12 N right; 12/4=3 m/s² right.

05. New evidence (reason)
Velocity 2 m/s left; net 15 N left; a 3 m/s² left.

06. Choose a method (mixed) *Review: Unit 1, Day 7; 12 course days later.*
Slowing down; change is 9 m/s east. Acceleration 2.25 m/s² east.

07. Exit check (exit)
Teacher looks for specific corrected method and evidence from new problem. Self-rating is reflection, not the teacher's mastery rating.

08. STRETCH: Apply it another way (stretch)
Example: zero net force means rest; counterexample steadily sliding puck. Zero net force means constant velocity.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 16; 3 course days later.*
Downslope weight 14.7 N; net 6 N down; a=2 m/s² down. Normal force≈25.46 N.

02. Explain a worked example (worked)
Directions match while speeding up and oppose while slowing in straight-line motion; motion direction alone is insufficient.

03. Find the first bad step (model)
v=(1-5)/2=-2 m/s; xf/t works for average velocity only if x i=0 and time is elapsed time.

04. Check the axis (practice)
Fx=-5 N; Fy=+8.66 N; angle is adjacent to north/y, west is negative.

05. Fresh recheck (reason)
Components (+6.93,-4.00) N; a=-3 m/s². Current velocity needed for motion direction.

06. Choose a method (mixed) *Review: Unit 1, Day 7; 12 course days later.*
First a=-6 m/s², speeding up. Then a=0, constant speed 35 m/s.

07. Exit check (exit)
Central 11 m; half-width 2 m; interval 9-13 m. Reflection must name a specific check, not just“double checked.”

08. STRETCH: Apply it another way (stretch)
Applied force 10 N; net 6 N; opposing force 4 N. Alternatives include incorrect moving mass, sensor calibration or nonconstant applied force; data needed.